to assess consistency and comparing agreement rates between LLM-human pairs versus human-human pairs, finding that GPT-4 achieves over 80% agreement with human preferences when biases are mitigated. Additionally, Wang et al. [67] provide comprehensive analysis of LLM-as-judge reliability across diverse evaluation scenarios. Critical disclosure requirements include documenting the evaluator model version, prompt template, temperature settings, and any detected biases in research reporting.

5) **Sampling-bias amplification**: This metric evaluates how the composition of in-context examples systematically affects exposure distribution for specific groups (e.g., language, genre, artist popularity tier). Lin et al. [89] demonstrate that recommendation models memorize item popularity in the principal spectrum and amplify this bias. For ICL contexts, bias amplification can be quantified by comparing group representation in recommendations versus catalog baselines, measured as relative popularity lift: $\Delta\text{Pop}(g) = \frac{\text{Pop}_{\text{rec}}(g) - \text{Pop}_{\text{catalog}}(g)}{\text{Pop}_{\text{catalog}}(g)}$, where positive values indicate over-representation. Evaluation should stratify across demographic dimensions and employ counterfactual seed sets to isolate causal effects of example composition.
6) **Order brittleness and query contradiction**: It quantifies recommendation instability arising from perturbations in example ordering or conflicts between explicit instructions and implicit example patterns. PromptSensiScore [194] measures sensitivity to prompt variants, finding that few-shot examples can alleviate instability with the most pronounced reduction in zero-to-one-shot transitions. Order brittleness can be measured via rank-biased overlap [168] across different orderings: $\text{OB} = 1 - \text{RBO}(R_{\pi_1}, R_{\pi_2})$, where π_1, π_2 represent different orderings of identical examples. Query contradiction is assessed by providing examples that violate stated constraints and measuring constraint adherence. When substantial instability is detected, fallback strategies include zero-shot prompting or explicit alignment scaffolds emphasizing instruction priority over example patterns.
7) **Privacy leakage and reproducibility**: These complementary metrics assess seed example memorization in outputs (privacy risk) and generation determinism (reproducibility). Carlini et al. [15] demonstrate that LLMs memorize training data, with extraction possible for verbatim sequences, particularly for repeated or distinctive content. In ICL contexts, leakage is measured as the proportion of seed example text surfacing in generated rationales or recommendations. For reproducibility, seed sets should be cryptographically hashed and logged alongside model version, temperature (with $T = 0$ ensuring determinism), and sampling parameters. Reproducibility enables systematic auditing when recommendation quality degrades, verified by regenerating recommendations with identical inputs and measuring output consistency.
8) **Implicit-feedback noise**: This metric quantifies how noisy implicit signals (skips, incomplete plays) in seed examples affect recommendation utility. Hu et al. [66] show that treating all implicit feedback uniformly degrades collaborative filtering performance, as signals like skips carry inherent ambiguity (dislike versus context inappropriateness). For ICL evaluation, one might compare recommendation quality across seed configurations: all interactions, positive-only signals, and confidence-weighted signals (e.g., high play completion rates). Measure utility via ranking metrics (nDCG@k) and diversity across configurations. When substantial quality degradation occurs with noisy seeds, filtering strategies should weight implicit signals by confidence indicators or exclude ambiguous interactions from ICL contexts.

3.2.3 Further Considerations on User and Item Modeling. Since the goal of automatically generated user profiles or item descriptions is to replace traditional user or item representation in a recommendation setting, the most straightaway methods of evaluating their success is through traditional recommendation metrics such as Precision, Hit rate, Recall, NDCG etc. In production, if users or content editors are allowed to edit user profiles or music content descriptors (e.g.,

playlist titles, album presentation), respectively, then the amount of correction required serves as a good indicator of the approach's ability to accurately represent users or items. However, most of the work on using LLMs to summarize users' preferences in a scrutable way has focused on domains such as books, movies, or e-commerce [56, 124, 193], whereas music presents its own unique challenges.

Given the context-length limitations of LLMs, it becomes crucial to devise effective sampling strategies for selecting items that best capture users' interests. Moreover, in the absence of explicit textual feedback, user or item modeling with LLMs must rely on item metadata, whose quality can vary substantially [100]. Bias is also quite present. In [144], authors show that profile quality, split into two dimensions (1) self-identification and (2) downstream recommendation, end up being biased towards types of music genres; for example, the tested LLMs generated better descriptions of taste when the users consumed metal music, which tends to be of American origin and are largely discussed on internet forums and magazines. More recent music tracks and content such as french Rap tended to be less generalizes, indicating that due to the corpora used to train LLMs, some content type end up being underrepresented. These differences raise fairness questions, if some users systematically don't recognize themselves, they're less likely to trust the system. Moreover, some items might be less reliably described by an LLM.

Moreover, due to the dynamic nature of conversations in LLMs, established datasets for offline evaluation are generally insufficient. Instead, users' explicit indication of their queries, emotions, intents, etc. need to be collected. Finally, assessing the contribution of advanced user models that integrate personas or psychological factors with music preferences in recommendation is considerably challenging, as disentangling the underlying factors is a non-trivial task.

LLMs and Long-Tail Items. LLMs struggle with *long-tail entities* [61, 69], a challenge that is particularly present in the music industry, where a large portion of the catalog consists of such lesser-known music artists or tracks. Recent research [69] has explored ways to address this limitation by introducing non-parametric knowledge into LLMs—either by augmenting prompts with relevant external information about the items or by injecting structured triplets from knowledge bases. Findings show that these approaches can significantly improve performance. In particular, prompting LLMs with knowledge base triples tends appears to work best, while combining both methods can substantially reduce hallucinations in the generated content. *However, when long-tail items correspond to new tracks or artists for which such external information is not yet available, either textually or within knowledge bases, this strategy remains infeasible, and alternative approaches still need to be explored. Evaluating performance on long-tail items is thus crucial, and at the very least, their impact should be systematically monitored.*

LLMs and Personas. Personalization lies at the core of recommender systems. In the context of LLMs, it could refer to their ability to represent different users or user groups; that is, to go beyond an "average" listener or music and adapt to diverse music preferences and behaviors. In other words, it reflects the model's steerability: its capacity to adjust its outputs based on the user. In practice, steerability can be achieved through prompting, fine-tuning, or activation steering [107]. Prompting remains the most common approach, as fine-tuning and activation methods often require access to model internals or significant computational resources. *However, Miehling et al. [107] show that prompt-based steerability remains highly challenging: it depends on model size, with larger models generally showing better steerability, and, more importantly, each model tends to be more responsive to specific subsets of persona dimensions by default. Given these findings, how can we fairly assess an LLM's ability to adapt while keeping in mind their inherent bias toward certain dimensions? Future research should measure not only overall quality but also how consistently models reflect individuals or groups, when used either in solutions or as evaluators.*

4 Conclusion

In summary, our work highlights how LLMs are already reshaping the development of MRS. Their ability to interpret and generate text enables richer user and item representations and supports interaction modalities that extend beyond traditional recommendation pipelines. At the same time, we reiterate the importance of carefully adapting evaluation practices to these emerging capabilities. The generative, context-dependent nature of LLM-based systems challenges long-standing assumptions in MRS evaluation, reinforcing the need for methodologies that remain reliable, transparent, and aligned with real user behavior. As LLMs continue to evolve, ensuring that evaluation frameworks evolve alongside them will be essential for maintaining rigor and trustworthiness in recommender-systems research.

We posit that this is an opportune moment for the community to take a step back and reflect on evaluation practices that more faithfully capture the underlying goals of music recommendation. Beyond "solving" datasets, we argue for frameworks that address what constitutes a good music recommendation and how systems can support long-term user preferences, fairness, and healthier consumption patterns. Given the nature of LLMs, traditional performance metrics based on predicting consumed items become increasingly problematic, both because of the opaque provenance of training data, which may overlap with popular recommendation datasets, and because such models can rely heavily on memorization to achieve high item-prediction scores. As we have described, the NLP community is comparatively more advanced in addressing these methodological concerns, and its practices can therefore serve as a foundation for future research to build upon.

We encourage future research on LLM-enhanced recommenders to remain attentive to the risks these models introduce and to design evaluation measures that genuinely reflect the goals and complexities of music recommendation. We expect our work to offer a grounded framework for navigating both the challenges of LLM usage and the inherent subtleties of musical preferences and interactions.

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